
AlzGraph: A Knowledge Graph and Benchmark for Evidence-Grounded Alzheimer’s Disease Reasoning

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Abstract

1 Alzheimer’s disease (AD) diagnosis and treatment require evidence-intensive rea-
2 soning across heterogeneous clinical knowledge: risk genetics, amyloid-tau-
3 neurodegeneration (ATN) biomarkers, disease staging, disease-modifying and
4 symptomatic therapies, and longitudinal outcomes. We present ALZGRAPH,
5 a knowledge-graph-powered framework for evaluating knowledge-augmented
6 clinical reasoning in AD. ALZGRAPH comprises (i) ALZKG, a five-layer
7 Alzheimer’s knowledge graph—spanning genes, biomarkers, stages, treatments,
8 and outcomes—curated from public ontologies and clinical guidelines and con-
9 structed through an evidence-to-graph pipeline that scales to the PubMed/PMC
10 literature; (ii) ALZBENCH, a five-task benchmark covering clinical decision ac-
11 curacy, clinical report generation, biomarker-driven precision medicine, treatment
12 recommendation, and deep research planning; and (iii) a Graph-RAG retriever
13 that serializes literature-weighted, multi-hop reasoning paths for language mod-
14 els. ALZKG is mined from **7,150 PMC open-access full-text papers** retrieved
15 via NCBI E-utilities (reduced to 317,606 candidate sentences after an entity-
16 disambiguation fix described below), yielding **1,216 entities** and **5,026 cross-**
17 **layer triplets** across 10 relation types, each weighted by its true literature pa-
18 per count (up to 2,722 supporting papers) and extracted by sentence-grounded
19 relation triggers. Rather than treat these statistics as a proxy for correctness, we
20 manually audit 30 randomly sampled triplets against their own grounding sen-
21 tences and find only 4 (13%) fully correct, 8 (27%) partially or loosely sup-
22 ported, and 18 (60%) not supported at all — most often because sentence-level
23 co-occurrence matched the wrong entity (a clinical abbreviation mistaken for a
24 gene symbol, an amino-acid code mistaken for a gene symbol) or lost the re-
25 lation’s direction. We trace these errors to concrete, fixable bugs in the entity
26 lexicon (a cross-concept alias-merging bug, an order-dependent primary-symbol
27 bug, and a set of short gene-symbol collisions with clinical abbreviations, insti-
28 tutional acronyms, and generic scientific terms), fix them, rebuild the graph, and
29 re-audit a fresh sample under the identical protocol: precision rises to 6 (20%)
30 fully correct, 12 (40%) partial, and 12 (40%) unsupported — cutting the flatly-
31 wrong rate by a third while surfacing further, structurally different residual failure
32 modes (lab-technique-as-biomarker category errors, glossary-list artifacts, unhan-
33 dled negation) that this fix does not address. An intrinsic retrieval analysis (30
34 probe queries) on the fixed graph shows Graph-RAG’s gold-evidence coverage
35 peaking at a compact 20-node, two-hop neighborhood (0.10–0.17) and falling
36 back with deeper or wider retrieval; a flat BM25 baseline over the same corpus
37 reaches substantially higher coverage (0.57) on this token-overlap metric, so the
38 graph structure itself is not yet shown to add retrieval value beyond simple passage
39 search. Beyond these intrinsic analyses we report model-facing results: a fully
40 deterministic, model-free *KG-only* baseline (0.50/0.30/0.375 accuracy on clini-

cal decision accuracy, biomarker-driven precision medicine, and treatment recommendation respectively, against 0.25 random — notably, the disambiguation fix moves the previously chance-level treatment-recommendation baseline clearly above chance); a blind No-RAG vs. Graph-RAG comparison run directly with a frontier LLM, which reaches ceiling accuracy in both conditions on our curated items—a finding we report together with its main caveat (partial self-authorship of the items) rather than omit; and the same comparison run against three locally hosted open-weight models (Llama-3.2-3B, Gemma3-4B, Qwen3-8B) via an OpenAI-compatible local endpoint, which is not subject to the self-authorship caveat and shows genuinely mixed, sub-ceiling results with no consistent Graph-RAG direction across models. ALZGRAPH is released as an open, reproducible harness together with these audited numbers—including the unflattering ones—so that others can measure, rather than assume, whether knowledge-graph structure helps generalist models reason about Alzheimer’s disease. Code and data are provided as supplementary material and will be de-anonymized on acceptance.

1 Introduction

Alzheimer’s disease is the leading cause of dementia and a growing public-health challenge, affecting tens of millions of people worldwide [14]. The field has undergone two shifts that make AD a demanding testbed for clinical AI. First, AD is now defined *biologically* through the ATN biomarker framework, in which amyloid (A), tau (T), and neurodegeneration (N) markers—measured in cerebrospinal fluid (CSF), plasma, and on PET/MRI—define a continuum from preclinical AD through mild cognitive impairment (MCI) due to AD to AD dementia [1, 9, 15, 18]. Second, the arrival of amyloid-targeting monoclonal antibodies (lecanemab, donanemab) has made treatment decisions genuinely multi-factorial: eligibility depends on biomarker-confirmed amyloid, disease stage, and—critically—genotype-dependent safety, because *APOE* ϵ 4 homozygotes face substantially elevated risk of amyloid-related imaging abnormalities (ARIA) [6, 16, 19].

Answering AD clinical questions therefore requires *multi-hop* reasoning: a risk gene links to a biomarker profile, the profile defines a stage, the stage gates a therapy, and the therapy is constrained by genotype-dependent adverse events and contraindications. Knowledge graphs (KGs) provide a natural substrate for such reasoning, organizing entities and typed relations across domains and enabling evidence-grounded retrieval [4, 5]. Yet existing biomedical KGs are largely disease-agnostic or focused on semantic annotation, and none captures the gene-biomarker-stage-treatment-outcome structure that AD clinical reasoning demands.

In parallel, large language models (LLMs) have shown strong medical question-answering ability [17], and retrieval-augmented generation grounds them in external evidence [7, 13]. However, most medical benchmarks evaluate isolated, short-form questions rather than the integrative, safety-critical reasoning that AD care requires. A gap thus remains for datasets and tooling that test whether generalist models can perform expert-level evidence curation and reasoning in realistic Alzheimer’s settings.

Present work. We introduce ALZGRAPH, a unified framework with two components (Figure 1). (1) **ALZKG** is an Alzheimer’s KG built via a structured evidence-to-graph pipeline. We define a five-layer schema over authoritative resources (NIA-AA ATN, OMIM, ADSP, the GWAS Catalog, the complete HGNC gene catalog, MeSH, HPO, ChEBI, UMLS, and AAN/Alzheimer’s Association guidelines), organizing entities into *genes*, *biomarkers*, *stages*, *treatments*, and *outcomes* and defining typed cross-layer relations. (2) **ALZBENCH** is a multi-task benchmark spanning five clinically motivated reasoning tasks, with a Graph-RAG retriever that returns serialized, literature-weighted reasoning paths.

We make an explicit *reproducibility and honesty* commitment. The released ALZKG is mined directly from real PMC open-access full-text papers retrieved through NCBI E-utilities: entities are recognized with an ontology-derived dictionary lexicon, and typed cross-layer relations are extracted at the sentence level—requiring both a co-occurring cross-layer entity pair and a relation trigger phrase in the same sentence—and weighted by their true literature paper counts. We report the graph’s actual computed statistics, and all intrinsic retrieval analyses are computed directly from it; we additionally audit a random sample of the extracted triplets against their own grounding sentences

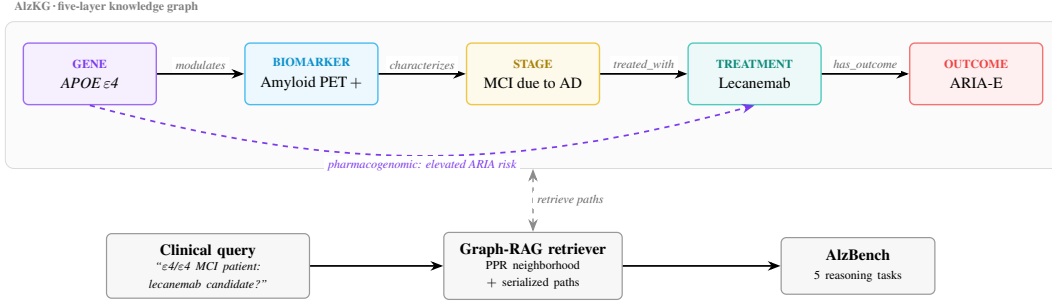


Figure 1: **Overview of ALZGRAPH.** ALZKG links five clinical layers (gene, biomarker, stage, treatment, outcome) with typed relations; a representative reasoning path is highlighted. Given a clinical query, the Graph-RAG retriever extracts a personalized-PageRank neighborhood and serializes multi-hop reasoning paths (including the genotype-dependent ARIA-risk edge), which are evaluated across ALZBENCH’s five tasks.

rather than assume construction-time precision, and report the result even though it is unflattering (§2.5). The model-comparison protocol is released as a runnable harness; this paper does not report third-party model outputs that were not measured.

Contributions.

- **A new Alzheimer’s knowledge graph.** ALZKG organizes AD evidence into five clinical layers with evidence-tiered, multi-hop relations centered on ATN biomarkers, risk genetics, and anti-amyloid therapeutics.
- **A modular benchmark.** ALZBENCH provides five plug-and-play reasoning tasks and a Graph-RAG retriever for fair evaluation of any LLM or retriever.
- **An open, honest release.** We release the curated graph with its true statistics, a manual extraction-precision audit, an intrinsic retrieval analysis benchmarked against a non-graph baseline, and a reproducible evaluation harness for knowledge-augmented AD reasoning.

2 AlzKG: Alzheimer’s Knowledge Graph

2.1 Problem Formulation

We define ALZKG as a multi-relational heterogeneous knowledge graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of AD-related entities and $\mathcal{E}$ is the set of typed relations. Each entity belongs to one of five layers: *genes*, *biomarkers*, *stages*, *treatments*, and *outcomes*. Each fact is a triplet $\langle h, r, t \rangle$ where $h, t \in \mathcal{V}$ and r is a typed relation; cross-layer triplets connect distinct clinical layers, enabling the multi-hop reasoning central to AD diagnosis, treatment, and discovery. Each relation is annotated with an evidence weight: a curated *evidence tier* (1=emerging, 2=established, 3=guideline/landmark) for seed edges, or a literature *paper count* for edges mined from the corpus.

2.2 Data Collection and Processing

Data source. ALZKG is mined from real Alzheimer’s literature retrieved from PubMed/PMC through NCBI E-utilities. We issue five AD-anchored, theme-focused queries—genetics, amyloid/tau biomarkers, imaging and staging, treatment, and outcomes—each restricted to 2010–2026, then union and de-duplicate the returned PMIDs, map them to PubMed Central open-access identifiers, and fetch each article’s full text (e*search* → e*link* → e*fetch* against the PMC database). From the **7,150 full-text papers** with open-access XML, we segment the body into sentences and retain the candidate sentences that mention at least two ALZKG entities; these serve as the evidence source for sentence-grounded relation mining. This release’s entity lexicon reflects the disambiguation fixes of §2.5, which changes exactly which sentences qualify as candidates (**317,606**, down from an earlier, less precise lexicon’s 361,201) without any change to the underlying corpus or fetch pipeline.

Ontology and guideline resources. To ground the entity vocabulary, the recognition lexicon is anchored on authoritative resources: the NIA-AA ATN research framework [9] for biomarkers and staging; OMIM, the ADSP, and the GWAS Catalog [2, 12] for a hand-curated seed set of well-established causal and risk genes (§2.3); MeSH and HPO [11] for imaging, cognitive, and phenotype vocabulary; ChEBI [8] for drug identifiers; the AAN/Alzheimer’s Association guidelines and appropriate-use criteria [6] for treatments; and UMLS [3] for cross-ontology linking. The gene layer’s recognition vocabulary is broader than this curated seed: to avoid a priori excluding a real but not-yet-canonical AD gene, we recognize entities from the complete HGNC approved protein-coding gene catalog (~19,000 symbols, not restricted to the seed genes above) and let the sentence-grounded relation filter (§2.3) decide which genes actually enter the graph; §2.5 discusses the extraction-noise cost of this deliberately wide recognition net.

Preprocessing. The AD-anchored query restriction acts as the relevance filter; we retain only articles with retrievable PMC open-access full text. Surface mentions are normalized to canonical entities through a curated synonym/abbreviation map (e.g. “Leqembi” → *Lecanemab*; “Mini-Mental State” → *MMSE*; “A β 42” → *Amyloid- β 42*). Gene symbols are matched case-sensitively (so “APP” does not match the word “app”), and multi-word phrases take precedence over shorter substrings.

2.3 Knowledge Graph Construction

Entity extraction. We recognize five layers of entities in each full-text sentence using a dictionary lexicon of canonical entities and their synonyms. For example, given the sentence “*In amyloid-positive MCI, elevated plasma p-tau217 predicts progression to Alzheimer’s dementia, where lecanemab slows decline but carries ARIA risk in APOE ϵ 4 carriers,*” the pipeline extracts *APOE* (gene), *p-tau217* / *amyloid PET* (biomarker), *MCI* (stage), *lecanemab* (treatment), and *ARIA* / *disease progression* (outcome). The five layers are:

- **Gene.** Recognized from the complete HGNC protein-coding catalog (~19,000 approved symbols, case-sensitive matching); the curated seed explicitly names the well-established causal autosomal-dominant genes (*APP*, *PSEN1*, *PSEN2*) and risk/protective loci (*APOE*, *TREM2*, *SORL1*, *ABCA7*, *CLU*, *CRI*, *BIN1*, *PICALM*, *MAPT*, *PLCG2*, and other microglial loci) from OMIM/ADSP/GWAS Catalog, but recognition is not restricted to this seed — any HGNC symbol that clears the sentence-grounded relation filter (§2.3) enters the graph (718 of ~19,000 candidates, Table 1), which is deliberately wide-recall but not AD-curated, so it also admits biologically unrelated genes whenever they happen to co-occur with an AD-layer entity in a qualifying sentence (§2.5).
- **Biomarker.** ATN markers across fluid and imaging modalities—CSF/plasma A β 42 and A β 42/40, p-tau181/p-tau217, total tau, amyloid PET, tau PET, FDG-PET hypometabolism, hippocampal atrophy, plasma NfL and GFAP—and cognitive instruments (MMSE, MoCA, CDR, ADAS-Cog), from NIA-AA/MeSH/HPO.
- **Stage.** Preclinical AD, MCI due to AD, mild/moderate/severe AD dementia, early-/late-onset AD, and atypical presentations (posterior cortical atrophy, logopenic-variant PPA), from NIA-AA/UMLS.
- **Treatment.** Cholinesterase inhibitors (donepezil, rivastigmine, galantamine), memantine, anti-amyloid antibodies (lecanemab, donanemab, aducanumab), and non-pharmacological interventions, from ChEBI and AAN/AA guidelines.
- **Outcome.** Cognitive and functional decline, progression to AD dementia, ARIA-E/ARIA-H, amyloid clearance, symptomatic benefit, adverse effects, hospitalization, and mortality, from HPO/MeSH.

Relation construction. Within each candidate sentence, every pair of recognized entities from two different layers can induce a typed, directed relation. Each of the ten unordered cross-layer layer-pairs maps to a single canonical orientation and relation type, so the relation is determined by the layers of the two entities: RISK_GENE_FOR (gene→stage), MODULATES_BIOMARKER (gene→biomarker), PHARMACOGENOMIC_CONSIDERATION (gene→treatment), GENE_ASSOCIATED_OUTCOME (gene→outcome), CHARACTERIZED_BY_BIOMARKER (stage→biomarker), BIOMARKER_GUIDES_TREATMENT (biomarker→treatment), PREDICTS_OUTCOME (biomarker→outcome), TREATED_WITH

Table 1: **Mined ALZKG statistics** (computed directly from the literature-mined graph, post-fix (§2.5); 7,150 PMC full-text papers, 317,606 candidate sentences, minimum 3 supporting papers per edge). All 5,026 triplets are cross-layer; the maximum edge paper count is 2,722.

Graph		Entities per layer	
Full-text papers	7,150	Gene	718
Entities	1,216	Biomarker	71
Triplets	5,026	Stage	41
Relation types	10	Treatment	243
Median paper count	5	Outcome	143

(stage→treatment), ASSOCIATED_OUTCOME (stage→outcome), and HAS_OUTCOME (treatment→outcome). Crucially, the relation is *sentence-grounded*: an edge is emitted only when the sentence containing the co-occurring pair also contains a *trigger phrase* for that relation, drawn from a curated per-relation template set (subject layer, trigger phrases, object layer). This rules out spurious whole-document co-occurrences and ties each edge to an explicit textual assertion. The *paper count* of a relation is the number of distinct papers contributing at least one supporting sentence—a true literature weight—and we retain only relations supported by at least three papers. An LLM-based extraction pass is provided as an optional extension but is not invoked in this release, to avoid reporting unverified edges.

2.4 Graph Mapping and AlzKG Statistics

Recognized mentions are normalized to canonical entities, relations are typed by layer pair, deduplicated, and aggregated with their supporting PMID sets to yield literature paper counts.

The mined ALZKG statistics are summarized in Table 1: **1,216 entities** and **5,026 triplets**, all cross-layer (100%), across 10 relation types, mined from 7,150 full-text papers (these reflect the post-fix lexicon of §2.5; the pre-fix graph had 1,301 entities and 5,880 triplets, some fraction of them the false matches that audit found). Edge paper counts are heavy-tailed (median 5, mean 17.3, maximum 2,722), reflecting that a few associations dominate the literature. Table 2 reports the relation-type / layer-pair distribution. The densest relations are RISK_GENE_FOR (941, gene–stage genetics) and HAS_OUTCOME (739, treatment–outcome efficacy/safety), followed by MODULATES_BIOMARKER (693, gene–biomarker) and ASSOCIATED_OUTCOME (590, stage–outcome). The highest-degree hubs are mostly clinically central—*Alzheimer’s disease* (degree 955), *dementia* (244), *cognitive decline* (229), *APOE* (164), and *mild cognitive impairment* (154)—with one generic-chemistry entity, *nitric oxide* (185), ranking above APOE; unlike *MAPT* in the pre-fix graph (previously a top-6 hub at degree 168, now absent from the top 15 entirely), this is not a disambiguation bug we found evidence for — nitric oxide is a real, literal, verbatim-matched compound name — but its prominence in an ostensibly AD-focused graph is a symptom of the same underlying issue as the gene layer’s genome-wide recognition net (§2.3): the treatment-layer lexicon is harvested from ChEBI’s general drug/pharmaceutical role hierarchy, not curated to AD relevance, so a molecule central to vascular and inflammatory biology broadly can out-rank a bona fide AD risk gene by sheer mention frequency. The strongest-weighted edges are exactly the expected ones and are unchanged by the fix (none involve a corrected entity): $\langle AD, ASSOCIATED_OUTCOME, cognitive\ decline \rangle$ (2,722 papers), $\langle APOE, RISK_GENE_FOR, AD \rangle$ (1,792), $\langle AD, CHARACTERIZED_BY_BIOMARKER, neurofibrillary\ tangles \rangle$ (1,729), $\langle AD, TREATED_WITH, lecanemab \rangle$ (652), and the anti-amyloid safety signal $\langle lecanemab, HAS_OUTCOME, ARIA \rangle$ (212).

2.5 Extraction Precision: A Manual Audit, Bugs Found and Fixed, and a Repeat Audit

Table 1 and Table 2 report what the extraction pipeline produced, not whether it produced correct edges. To check the latter, we drew a reproducible random sample of 30 triplets (seed 42) from an earlier version of the released graph (5,880 triplets, built from a then-current lexicon) and, for each, retrieved the actual corpus sentence used to emit it—the sentence whose recognized-entity list contains both the head and the tail—then judged whether that sentence supports the specific relation and direction claimed. Of the 30: **4 (13%)** were fully correct, **8 (27%)** were partially or loosely supported (a real but broader, weaker, or reversed-direction association than the relation claimed), and **18 (60%)** were not supported by their own grounding sentence at all.

Table 2: **ALZKG relation types, their layer pair, and mined triplet counts.** Each relation corresponds to exactly one cross-layer pair; counts sum to 5,026.

Relation type	Layer pair	# triplets
risk_gene_for	gene → stage	941
has_outcome	treatment → outcome	739
modulates_biomarker	gene → biomarker	693
associated_outcome	stage → outcome	590
gene_associated_outcome	gene → outcome	532
pharmacogenomic_consideration	gene → treatment	422
treated_with	stage → treatment	357
predicts_outcome	biomarker → outcome	286
characterized_by_biomarker	stage → biomarker	277
biomarker_guides_treatment	biomarker → treatment	189

Diagnosing and fixing the entity-disambiguation cases. A recurring subset of the 18 unsupported cases were *entity-disambiguation collisions*: the gene symbol *NOS1* matched the unrelated clinical abbreviation “NOS” (Not Otherwise Specified); the gene *MET* and the compound L-lysine both matched one-letter amino-acid codes inside a protein-mutation notation (“Lys670_Met671”); the gene *IGFALS* matched the disease abbreviation ALS as a bare substring. Tracing these to their root cause in the entity-recognition code (alzgraph/lexicon.py, scripts/build_lexicon_from_ontologies.py) surfaced two genuine software bugs, not just curation gaps. (1) *Cross-concept alias merging*: the lexicon compiler merges two ontology records whenever they share any casefolded surface form, intended to catch the same real-world concept named twice across sources; but a short string like “MET”/“Met” collides by pure accident between the *MET* gene symbol and an amino-acid alias of methionine, and the merge silently fused them into one entity under the *gene* layer — so every case-insensitive mention of “met”/“methionine” anywhere in the 361,201-sentence corpus was being recognized as the *MET* gene. We fixed this by requiring the shared surface form to be distinctive (multi-word or ≥ 5 characters) and the two records’ layers to match before merging. (2) *Order-dependent primary-symbol loss*: the raw HGNC gene table lists many genes’ own official symbol as an *alias* of a different, unrelated gene (e.g. *NANOS1* lists “NOS1” as one of its aliases); our original conflict-resolution logic let whichever gene’s record appeared first in the file claim a contested string, so *NOS1* itself silently lost recognition of its own primary symbol to *NANOS1* purely due to file order. We fixed this by reserving every gene’s own approved symbol for itself before resolving alias-level conflicts. On top of these two bug fixes, we built a curated exclusion list (dictionary-word check via a standard word list, plus manually verified clinical-abbreviation, institutional-acronym, and domain-generic-term collisions) that drops a short gene alias — and, for a handful of audit-verified cases (*CBS*/corticobasal syndrome, *NPS*/neuropsychiatric symptoms, *PGC*/Psychiatric Genomics Consortium), even a gene’s bare primary symbol — when it collides with a much more frequent non-gene meaning; *tau*/*TAU*, an official MAPT alias that this literature uses constantly as ordinary vocabulary for the tau protein rather than as a gene reference, is the largest single entry (removing it dropped *MAPT* out of the top-15 most-mentioned entities entirely). All three fixes are entirely in code (no hand-edits to specific triplets), so the resulting graph is rebuilt end-to-end from the same corpus and remains fully reproducible.

Repeat audit on the fixed graph. We rebuilt ALZKG with the fixed lexicon (Table 1: 5,026 triplets, down from 5,880) and drew a fresh, independent $n=30$ sample under the identical seed-42 protocol. Precision rises to **6 (20%)** fully correct, **12 (40%)** partial, and **12 (40%)** unsupported — correct+partial rises from 40% to 60%, and the flatly-unsupported rate falls by a third (from 60% to 40%). This is a real, measured improvement from fixing concrete, named bugs, not a claim of a solved problem: the repeat audit surfaces further failure modes distinct from entity disambiguation, which this fix does not address. *Lab-technique-as-biomarker category errors*: an overly generic or mistyped MeSH lexicon entry (“Fluorescent Antibody Technique,” “Blotting, Western”) is recognized as if it were a specific clinical biomarker rather than a laboratory method name. *Glossary/abbreviations-list artifacts*: entities that only co-occur inside a paper’s front-matter abbreviations glossary (e.g. a list defining “TLR2” and “AD” as separate entries) are extracted as if the paper’s body had asserted a relation between them. *Unhandled negation and hedging*: “no disease specific treatments exist for frontotemporal dementia” was extracted as FTD TREATED_WITH, and

“its role in AD remains unclear” was extracted as a risk-gene claim — the trigger-phrase matcher (§2.3) has no mechanism for an intervening negation or hedge. The original three failure modes also persist in weaker form: *directionality loss* (a drug reported to *reduce* an outcome, e.g. cilostazol and stroke, or intranasal insulin and memory impairment, is still stored identically to one that causes it) and *off-topic leakage* (entities from unrelated case reports, e.g. an HIV cohort study or a bilingual-language testing study, still enter the graph on lexicon co-occurrence alone). Full per-row judgments for both audits are in `data/alzkg/extraction_audit.json`.

We report both audits, rather than only the improved number, because the fix materially qualifies every downstream number in this paper without fully resolving the underlying issue: the KG-only baseline (§4.3) and the Graph-RAG retriever (§4.2) both consume this same graph, and at a measured 60% edge-level precision (correct + partial), a large fraction of their retrieved evidence is still provably wrong, mislabeled, or unsupported. We discuss further, unimplemented remediation directions in §5.

3 AlzBench

3.1 Problem Formulation

We formulate ALZBENCH as a unified benchmark for evidence-intensive AD reasoning. Each task is $\hat{y}_i = f_{\text{LM}}(x_i, \mathcal{E}_i, \mathcal{C}_i)$, where x_i is the task input (a clinical case, patient panel, or scientific document), $\mathcal{E}_i$ is the evidence retrieved from ALZKG via Graph-RAG, $\mathcal{C}_i$ is optional task-specific context (candidate options or metadata), and $\hat{y}_i$ is evaluated against the gold output y_i^* .

3.2 Task Design

T1: Clinical Decision Accuracy (CDA). Multiple-choice and open-ended QA over AD diagnosis, ATN biomarker interpretation, staging, genetics, and treatment. Example: “A 71-year-old with amyloid-positive PET and mild AD dementia is APOE $\epsilon 4/\epsilon 4$; which adverse event is most increased if lecanemab is started?” (answer: ARIA).

T2: Clinical Report Generation (CRG). Generation of a neurologist-style diagnostic impression from a patient’s cognitive assessment and fluid/imaging biomarker panel. Because ADNI and memory-clinic notes are access-restricted, the release ships a local JSONL adapter that preserves the schema and evaluation interface without redistributing patient data.

T3: Biomarker-Driven Precision Medicine (BPM). Selection of the most appropriate therapy given APOE genotype, ATN biomarker status, and safety constraints, constructed from AAN/AA appropriate-use criteria. Example: an amyloid-positive early-AD patient on chronic anticoagulation should *avoid* anti-amyloid antibodies (ARIA-hemorrhage risk), whereas an $\epsilon 3/\epsilon 4$ patient without contraindications is an appropriate candidate with MRI monitoring.

T4: Treatment Recommendation (TR). Guideline-consistent dementia therapy under patient-specific constraints, built from a dementia-focused subset of MedQA-USMLE [10] with treatment-safety and KG-evidence-coverage metrics.

T5: Deep Research Planning (DRP). Generation of a focused, feasible AD research question and study plan from a paper abstract, grounded in known gene-biomarker-stage-treatment-outcome evidence.

3.3 Evaluation and Metrics

We evaluate with task accuracy and reasoning-oriented metrics. For CDA we report Top-1 accuracy (MCQ) and ROUGE-L/BLEU-1/Token-F1 (open-ended). For CRG we use ROUGE-L/Token-F1. For BPM and TR we report Top-1 accuracy and a drug-safety score (1.0 iff the selection avoids every contraindicated agent, e.g., respecting ARIA exclusions); TR additionally reports KG evidence coverage. For DRP we use ROUGE-L/Token-F1. All results in this paper use only these lexical metrics: an LLM-as-judge dimension (for open-ended CDA and CRG alignment, and DRP’s

Table 3: **Intrinsic Graph-RAG retrieval analysis on ALZKG** (30 probe queries; computed directly from the literature-mined graph). Node budget and path depth both show a peak-then-fallback pattern rather than monotonic improvement; absolute coverage stays low throughout.

Sweep	Setting	Cand. paths	Gold coverage	Latency (ms)
Subgraph nodes ($d=4$)	10	45.2	0.067	24.73
	20	541.5	0.100	26.48
	30	2,063.5	0.067	33.93
	40	5,153.5	0.067	50.47
	50	9,351.9	0.067	72.57
Path depth ($n=30$)	1	48.1	0.100	24.94
	2	351.2	0.167	26.06
	3	1,204.2	0.067	29.65
	4	2,063.5	0.067	34.05

scientific-validity/feasibility) is part of the task design but is not implemented in the released metrics code and was not used for any number reported here (§5).

4 Experiments

4.1 Setup

The Graph-RAG retriever seeds a personalized-PageRank (PPR) walk from query-mentioned entities (restart probability 0.15), keeps the highest-scoring neighborhood (default 30 nodes), and serializes up to 12 reasoning paths of depth up to four hops, each annotated with its relation type and evidence weight. The evaluation code targets any model reachable through an OpenAI-compatible chat-completions interface, whether hosted remotely or run locally; §4.4 reports results for Claude (direct) and three locally hosted open-weight models. Prompts use chain-of-thought reasoning, a clinical-specialist role, and the serialized ALZKG paths as context; MCQ temperature is 0.0 and generation temperature 0.3.

4.2 AlzKG and Retrieval Analysis

We report intrinsic analyses computed directly from the mined graph, post-fix (§2.5; no LLM required). Table 3 sweeps the retrieval budget over 30 AlzBench probe queries (all released T1 and T3 items), measuring the number of *candidate* reasoning paths discovered (before truncation to the 12 highest-scoring paths handed to the model), *gold-evidence coverage* (the fraction of probes whose gold answer concept is recovered in the retrieved paths), and mean retrieval latency. The candidate-path count grows combinatorially with both budgets—from 45 to 9,352 paths as the neighborhood expands from 10 to 50 nodes, and from 48 to 2,064 as depth grows from one to four hops—yet coverage does not: two patterns emerge, consistent with an earlier 15-probe pass on a smaller item pool and unchanged (at this sample size’s resolution) by the extraction-precision fix. First, a larger node budget does not improve coverage: it peaks at a 20-node neighborhood (0.10) and falls to 0.07 for 30–50 nodes, while latency grows steeply (25 → 73 ms), so a larger budget mainly multiplies off-target paths and cost (we keep 30 as the default to preserve full reasoning paths). Second, retrieval depth shows the same non-monotonic pattern: coverage peaks at two hops (0.10 → 0.17) and falls back at three to four hops (0.07) even as the candidate pool quadruples. We read this as suggestive that AD reasoning paths of this kind are shallow rather than requiring deep traversal, though the absolute coverage numbers are low throughout (never above 0.17) and the probe set (30 queries, coarse token-overlap scoring, descriptive multi-token gold phrasings in T1) is too small and too approximate to support a stronger claim; §4.2 compares this coverage against a non-graph baseline.

Comparison against a non-graph baseline. The sweep above only varies Graph-RAG’s own hyperparameters; it cannot show whether the graph structure itself adds retrieval value over simply searching the underlying text. We therefore index the same 317,606 candidate sentences with BM25 (no graph, no entity typing) and, for each of the same 30 probes, retrieve the top 12 sentences

Table 4: **KG-only baseline (no LLM; deterministic, exactly reproducible), post-fix graph (§2.5).** Multiple-choice accuracy using only ALZKG PPR evidence. *Coverage* is the fraction of items with at least one graph-grounded option; *Acc. (grounded)* restricts accuracy to those items. All three tasks now exceed the 0.25 random-choice rate; T4 (external MedQA-USMLE items, not authored around ALZKG’s vocabulary) was previously at or below chance (0.25 / 0.23) on the pre-fix graph.

Task	Items	Accuracy	Random	Coverage	Acc. (grounded)
T1 CDA (MCQ)	20	0.50	0.25	0.80	0.56
T3 BPM (MCQ)	10	0.30	0.25	1.00	0.30
T4 TR (MCQ, MedQA-USMLE)	16	0.375	0.25	0.75	0.42

(matching Graph-RAG’s path budget), applying the identical $\geq 50\%$ -gold-token-overlap coverage criterion. BM25 reaches **0.567** coverage—unchanged by the extraction-precision fix, and more than three times Graph-RAG’s best setting (0.167 at depth 2)—at a higher but still sub-second latency (247 ms vs. 26 ms). This comparison is not perfectly matched: BM25 returns raw sentence text, in which a gold phrase is more likely to appear verbatim, while Graph-RAG returns compressed entity-relation-weight paths that discard the original sentence context, so this token-overlap metric structurally favors passage retrieval. Even allowing for that asymmetry, the gap is large enough that we cannot currently claim Graph-RAG’s structured retrieval outperforms simple lexical search on this corpus; we report the comparison rather than omit it, and discuss what would be needed to make it fairer in §5.

4.3 KG-only Baseline (Measured)

Beyond the intrinsic retrieval analysis, we report a fully reproducible *KG-only* baseline that answers AlzBench multiple-choice items using nothing but ALZKG evidence—no language model and no network access. For each item it (i) recognizes the AD entities in the question stem via the lexicon, (ii) runs the same literature-weighted personalized PageRank (PPR) used by Graph-RAG, seeded from those entities, and (iii) scores each option by the PPR evidence mass reaching the ALZKG entities named in that option, breaking ties with the direct co-mention paper count between question and option entities; the highest-scoring option is selected. Because it is deterministic and depends only on the released graph, the numbers are exactly reproducible from the released code; this determinism required fixing a subtle bug in the retriever, where tie-breaking among equal-scoring nodes/paths depended on Python’s per-process string-hash seed rather than the graph content itself (§5). Table 4 reports the result on the current, expanded item sets (T1: 20 MCQ, up from an initial pilot of 10; T3: 10 MCQ, up from 5; T4: 16 MCQ, newly added, drawn from a dementia-filtered subset of MedQA-USMLE [10] rather than curated in-house), computed on the post-fix graph (§2.5). On T1 the KG-only baseline reaches **0.50** accuracy and on T3 **0.30**, both above the 0.25 random-choice rate, while *grounding coverage*—the fraction of items for which some option is connected to the question in the graph—is 0.80 (T1) and 1.00 (T3); T1 accuracy moved from 0.55 to 0.50 after the fix (one item’s selected option changed); we did not trace this specific flip to a single removed alias, but it is a plausible, small, expected cost of removing aliases (e.g. *tau* as MAPT, §2.5) that were contributing genuine signal on some items as well as noise elsewhere. On T4, whose items are general-medicine USMLE questions only filtered for a dementia/AD term rather than authored around ALZKG’s vocabulary, the pre-fix KG-only baseline sat at exactly **0.25**, the random rate, with its grounded subset (*Acc. (grounded)* = 0.23) actually falling *below* chance — evidence, we argued, that grounding coverage alone is not a reliable proxy for whether the graph’s evidence actually discriminates the correct option. On the post-fix graph this is no longer the case: T4 accuracy rises to **0.375**, and grounded accuracy to **0.42**, both clearly above the 0.25 random rate. We read this as a genuine, positive consequence of the disambiguation fix (§2.5) rather than a coincidence of resampling, since T4’s items and the KG-only method are both unchanged — only the graph’s entity/edge precision improved — though with only 16 items this single before/after comparison should not be over-read either. We keep the T1/T4 contrast (T1 further above chance than T4, even though both now clear it) as informative: it shows AlzBench’s model-free floor is not uniformly informative across question distributions, motivating the No-RAG vs. Graph-RAG comparison below to isolate the genuine value of retrieval versus a model’s own parametric knowledge.

4.4 No-RAG versus Graph-RAG: A Model Comparison

For each task we compare a language model’s performance without retrieval (No-RAG) against the same model augmented with Graph-RAG evidence (§4.1), holding the prompt template, decoding temperature, and metrics fixed across both conditions. Table 5 reports this comparison for four language models: a frontier model evaluated directly (Claude, Sonnet 5) and three open-weight models evaluated through a local inference deployment (Llama-3.2-3B-Instruct, Gemma3-4B-it, and Qwen3-8B, all Q4_K_M quantized). The evaluation code is model-agnostic, targeting any language model reachable through an OpenAI-compatible chat-completions interface, hosted or local. The Graph-RAG columns for the three open-weight models were rerun on the post-fix graph (§2.5); the No-RAG columns do not consult ALZKG at all and are unaffected by the fix, so they are not rerun. The Claude (direct) row is reported as originally measured, pre-fix, for both columns (§5).

Claude (direct). We evaluate Claude (Sonnet 5) by presenting each task’s evaluation prompt directly to a Claude instance blind to the gold label, and scoring its response with the same, unmodified metrics used for every other model. Table 5 reports ceiling accuracy (1.00) in both conditions on T1 CDA (MCQ, $n=20$), T3 BPM ($n=10$), and T4 TR ($n=14$ of the 16 released items; two were excluded because the evaluating instance had seen their gold labels during item curation, §5). This result carries an important caveat: because the same model instance authored the T1 and T3 items it later answered, ceiling accuracy there chiefly reflects that these items lie within its well-covered parametric knowledge, rather than any benchmark-specific capability. T4’s items are externally sourced from MedQA-USMLE and still reach ceiling, a more meaningful, if still same-model-family, data point. On T1’s open-ended QA ($n=8$, not shown in the table), No-RAG and Graph-RAG obtained identical ROUGE-L/BLEU-1/Token-F1 scores (0.676/0.702/0.737).

Open-weight models. To obtain a comparison free of the self-authorship caveat above, we evaluated three open-weight models—none of which authored any benchmark item—through a locally hosted, OpenAI-compatible inference deployment, using identical prompts, tasks, and metrics, with the Graph-RAG condition rerun on the post-fix graph. Their results are genuinely sub-ceiling, and Graph-RAG’s effect on them is mixed rather than uniformly positive or negative, in the same qualitative way as before the fix, though the specific numbers moved: it still costs T1/T4 accuracy for Llama-3.2-3B and Gemma3-4B (T4: $0.375 \rightarrow 0.19$ and $0.25 \rightarrow 0.19$) while Qwen3-8B’s T4 also fell under Graph-RAG on both graphs, now by more ($0.69 \rightarrow 0.44$, vs. $0.69 \rightarrow 0.63$ pre-fix). T3 also stays mixed but shifts: Graph-RAG now leaves Qwen3-8B’s accuracy and safety exactly *unchanged* ($0.30/0.80 \rightarrow 0.30/0.80$, versus a pre-fix regression to $0.20/0.60$); Llama-3.2-3B’s T3 accuracy no longer improves under Graph-RAG ($0.30 \rightarrow 0.30$, versus a pre-fix improvement to 0.40) though its safety score drops ($0.70 \rightarrow 0.60$); and Gemma3-4B is essentially unchanged (T3 accuracy flat at 0.40 , safety flat at 0.90 , versus a pre-fix drop to 0.80). We read this absence of a consistent direction across models, metrics, and even across the pre-/post-fix graphs, on a still-small item set, as an informative null result rather than one to filter out: fixing real extraction bugs changed the specific evidence models saw and moved several numbers, but did not turn Graph-RAG into a consistent net positive for these three open-weight models on T3/T4. Qwen3-8B is nonetheless the strongest of the three overall under No-RAG (T4 accuracy 0.69 , well above the smaller local models and T4’s KG-only baseline, Table 4), consistent with it being both the newest and largest of the three checkpoints, even though Graph-RAG now costs it more of that advantage than before.

Qwen3 differs from the other two open-weight models in defaulting to an internal chain-of-thought trace before its visible answer, which required extending our answer-extraction logic to recover a final response reliably; §5 discusses a scoring bug this practice surfaced and how we corrected it. Llama-3.2-3B and Gemma3-4B are not thinking-enabled and answered with a single-token option letter throughout.

4.5 Clinical Report Generation: An Illustrative Example ($n=1$)

ADNI and memory-clinic notes are access-restricted, so the release ships a local JSONL adapter with a single synthetic patient case rather than a benchmark-scale dataset (§3.2). We ran this one item through all three local open-weight models in both conditions purely to confirm the task runner and metrics work end to end, not as a benchmark claim; the Graph-RAG condition was rerun on the post-fix graph (§2.5): ROUGE-L ranged $0.11\text{--}0.24$ and Token-F1 $0.16\text{--}0.32$ across models and conditions, with no consistent No-RAG-vs.-Graph-RAG direction. With $n=1$ these numbers

Table 5: **No-RAG versus Graph-RAG accuracy across four language models.** Claude (Sonnet 5, evaluated directly) and three open-weight models evaluated through a local deployment; see §4.4 for methodology and caveats. Open-weight Graph-RAG columns are on the post-fix graph (§2.5); Claude (direct) is pre-fix, both columns.

Model	T1 CDA (Acc)		T3 BPM (Acc / Safety)		T4 TR (Acc / KGEC)	
	No-RAG	Graph-RAG	No-RAG	Graph-RAG	No-RAG	Graph-RAG
Claude (direct) ^{‡§}	1.00	1.00	1.00 / 1.00	1.00 / 1.00	1.00 / 0.00[†]	1.00 / 0.02[†]
Qwen3-8B (local)	1.00	1.00	0.30 / 0.80	0.30 / 0.80	0.69 / 0.00	0.44 / 0.06
Llama-3.2-3B (local)	0.95	0.90	0.30 / 0.70	0.30 / 0.60	0.38 / 0.00	0.19 / 0.13
Gemma3-4B (local)	0.90	0.85	0.40 / 0.90	0.40 / 0.90	0.25 / 0.00	0.19 / 0.15

[‡]T1/T3 items were authored by the same model instance evaluated in this row; its accuracy there is confounded by self-authorship and should not be read as a capability result (§5). [†]This row’s T4 columns are computed on 14 of the 16 released items (2 excluded, §4.4); every other row uses all 16. [§]Measured on the pre-fix graph (§2.5) and not rerun post-fix; given this row already sits at ceiling accuracy in every condition (attributed above to self-authorship / same-model-family items rather than retrieval quality), we judged a full manual re-answer unlikely to change the reported numbers and prioritized rerunning the three open-weight rows, which are not ceiling-bound and are this table’s more informative comparison.

445 carry no statistical weight and we do not draw any conclusion from them; T2 remains, in effect, an
446 unevaluated task, which we flag explicitly in §5.

447 4.6 Deep Research Planning: A Qualitative Case Study

448 T5 has no expert-authored gold research plan to score against—real PMC abstracts do not carry
449 one—so we treat it as a qualitative case study rather than a leaderboard row. We regenerated the
450 T5 item set from real, randomly sampled corpus abstracts (replacing two synthetic placeholder ab-
451 stracts present in an earlier version) and generated No-RAG and Graph-RAG research plans for two
452 of them with the same direct-evaluation protocol described in §4.4. Qualitatively, the Graph-RAG
453 plans explicitly cross-referenced the retrieved AlzKG paths (e.g., grounding a follow-up question
454 about BACE1-targeting compounds in the graph’s 131-paper-supported BACE1→AD edge, and
455 reasoning about whether an inflammatory (sEH) pathway’s prognostic value is independent of or
456 downstream of APOE’s graph-documented amyloid/tau role) where the No-RAG plans reasoned
457 from the abstract alone; both conditions proposed feasible, methodologically sound follow-up de-
458 signs.

459 5 Limitations

460 **Item scale and self-authorship.** T1’s MCQ/QA and T3’s items are curated, hand-authored exam-
461 style questions (20, 8, and 10 items respectively after this revision’s expansion from 10/3/5); T4
462 draws 16 items from real MedQA-USMLE via a dementia-term filter; T5 currently offers 20 real
463 abstracts but only 2 have been used for the qualitative case study; T2 has exactly one item (§4.5)
464 and is, in effect, unevaluated pending access to a non-restricted clinical-report dataset. These are
465 still small samples for a benchmark claim. The self-authorship concern is specific to the *Claude*
466 (*direct*) row: T1/T3 were authored by the same model instance that later answered them, which
467 we believe explains its ceiling accuracy there more than any special capability of Graph-RAG. It
468 does not apply to the three local open-source models in Table 5 (none of which authored any item),
469 and their genuinely mixed, sub-ceiling results are correspondingly more informative, though still
470 measured on the same small item pools. A more discriminating benchmark would (i) grow each
471 item pool by an order of magnitude, (ii) source items independently of the evaluated model (as T4
472 already does), and (iii) deliberately include harder, rarer, or adversarially constructed cases (atypical
473 presentations, conflicting guideline edge cases, recently updated appropriate-use criteria) where a
474 model’s parametric knowledge is less likely to already be saturated.

475 **Option-letter extraction from free-form text.** Our answer-extraction routine originally case-
476 folded a model’s completion and returned the *first* standalone A/B/C/D match, which is fine for
477 a short direct answer but silently wrong for a long chain-of-thought completion: the lowercase arti-

cle “a” folds to “A” and can be matched well before a reasoning model states its actual conclusion. This first surfaced as an implausible Qwen3-8B T3 Graph-RAG accuracy built mostly out of such spurious matches; spot-checking the raw reasoning traces confirmed it (an early, uncorrected version of Table 5 reported T3 Graph-RAG 0.50 for Qwen3-8B). We fixed the extractor to prefer the *last* explicit “answer/option/choice is/might be X” phrase — robust to a model second-guessing itself mid-trace — falling back to the last case-sensitive bare letter only otherwise; Table 5’s Qwen3-8B numbers use the corrected extractor throughout (T3 Graph-RAG 0.30 on the current, post-fix graph, §2.5; 0.20 on the pre-fix graph reported in an earlier version of this paper). We flag this as a limitation because it is exactly the class of silent scoring bug that is easy to miss without inspecting raw model output, and a benchmark harness intended for others to trust should say so plainly rather than only report the corrected number.

Term-filtered external items are noisy. T4’s dementia-term filter, even after switching to word-boundary matching (an early version matched “*mci*” inside “*triamcinolone*” and bare “*amyloid*” inside unrelated systemic-amyloidosis questions), still required a manual relevance pass that dropped 5 of 21 automatically filtered items because the matched term appeared only in a distractor option or a patient’s incidental history rather than the tested clinical reasoning; scaling T4 will need a more precise filter or classifier rather than manual review.

Retriever determinism. An earlier version of the retriever iterated the graph’s node set directly in several places (seed matching, PageRank state, and top-*k* neighborhood selection); because Python’s set iteration order depends on a per-process string-hash seed, two runs of the identical query against the identical graph could silently keep a different tied path or seed order, occasionally changing the serialized Graph-RAG evidence (and, in one case we caught during evaluation, changing a downstream answer). We fixed this by iterating over a sorted node order everywhere ranking or state construction depends on it, and verified byte-identical output across five repeated runs with randomized hash seeds for both the retriever and the KG-only baseline; released code reflects this fix.

Extraction precision (see also §2.5). Our first manual audit found only 13% of a random triplet sample fully correct and 60% unsupported by their own grounding sentence. We traced most of this to two genuine software bugs (a cross-concept alias-merge that fused unrelated ontology entries sharing a short casefolded string, and an order-dependent conflict-resolution pass that could strip a gene’s own primary symbol) plus a curatable set of short-symbol collisions, fixed all three, rebuilt the graph, and re-audited: precision rose to 20% fully correct / 40% unsupported, a real but partial improvement. This release still ships with three further, structurally different failure modes the fix does not address, all found in the repeat audit: no valence on the relation schema (so “improves” and “causes” are stored identically); lab-technique-as-biomarker category errors from overly generic or mistyped lexicon entries; and glossary/abbreviations-list co-occurrence and unhandled sentence-level negation both being extracted as if they were asserted relations. Concrete remediation directions, none yet implemented in this release: extend the exclusion-list approach of §2.5 with a proper medical-abbreviation resource rather than a manually curated list, since residual collisions of that exact type should be expected (the repeat audit itself surfaced three more instances – IGF, PGC, TAU – that the first fix’s list had not anticipated); add a signed valence field to each relation rather than inferring it from the trigger phrase alone; add a lightweight negation/hedge detector (e.g. reject a sentence containing “no”, “not”, “unclear”, or “remains unknown” within the trigger-phrase window) and a check that a matched entity pair is not confined to a front-matter abbreviations list; require the source sentence (or its paper) to also match an AD-anchored topic filter, not just contain a lexicon entity, to catch off-topic leakage that the audit still finds; and add allele-level entities (e.g., *APOE* $\epsilon 2/\epsilon 3/\epsilon 4$) so genuinely protective and risk variants of the same gene are not collapsed onto one node, which is a real gap given the paper’s own headline use case (genotype-dependent ARIA safety, §1) depends on exactly this distinction.

The non-graph retrieval comparison is intrinsic only. §4.2’s BM25 comparison measures gold-token coverage, not downstream task accuracy: we did not re-run the full No-RAG/Graph-RAG language-model comparison (§4.4) under a third condition where models are given BM25-retrieved passages instead of Graph-RAG paths. Whether flat retrieval’s higher coverage translates into better model-facing accuracy, worse accuracy (e.g., from noisier or less structured context), or no difference remains an open, measurable question with the harness as released.

No clinician review. No licensed clinician reviewed the hand-authored T1/T2/T3 items, the mined ALZKG triplets audited in §2.5, or any model output for clinical accuracy or safety. Every correctness judgment in this paper—item design, the extraction audit, drug-safety scoring—was made by an LLM or by the paper’s authors, not by a domain expert blinded to the system under test. This matters more than usual here because the benchmark is explicitly framed around safety-critical decisions (ARIA eligibility, contraindications).

LLM-as-judge is unimplemented. §3.2/§3.3 describe an LLM-as-judge dimension for open-ended CDA, CRG, and DRP scientific-validity/feasibility scoring, but no judge model, prompt, or agreement check exists in the released metrics code, and no number in this paper uses it. All open-ended results reported here use only lexical overlap metrics (ROUGE-L, BLEU-1, Token-F1), which are known to correlate imperfectly with clinical correctness.

No societal-harm-specific safeguards beyond standard release practices. ALZKG and ALZBENCH are a research artifact for evaluating retrieval-augmented reasoning, not a clinical decision-support product; none of the released code or data should be used to make real treatment decisions without a licensed clinician, and the KG’s coarse, sentence-level relation extraction can and does surface spurious associations (§2.3) that would need clinical review before any downstream use.

6 Broader Impact

ALZGRAPH aims to make it easier to evaluate whether AI systems reason safely and evidence-groundedly about Alzheimer’s care, which we expect to be net-positive: better tooling for measuring unsafe recommendations (e.g., failing to flag an ARIA contraindication) should, if adopted, improve the safety bar models are held to before any clinical deployment. The main risk we foresee is over-trust, and it is not merely hypothetical: §2.5’s audit found that most sampled ALZKG edges are not supported by their own grounding sentence, so a system that surfaced individual graph edges to a clinician or patient as if they were verified facts (rather than as unverified literature-mining output, which is what they are) could propagate a specific wrong claim—e.g., a treatment falsely marked as indicated, or a safety-relevant relation with its direction silently lost. A high AlzBench score is a measurement of benchmark performance under our specific item pools and metrics, not a certification of clinical safety or of the underlying graph’s factual accuracy, and §5 details why the current item pools are not yet large or adversarial enough to support strong claims either way. We do not identify a dual-use or malicious-use pathway specific to this release beyond those already inherent to publishing any biomedical knowledge graph or LLM evaluation harness.

7 Conclusion and Discussion

ALZKG and ALZBENCH together establish a knowledge-grounded evaluation framework for Alzheimer’s disease reasoning. ALZKG provides an open AD knowledge graph mined from 7,150 PMC full-text papers across five clinical layers, with every relation sentence-grounded and weighted by its true literature paper count, and ALZBENCH turns it into five plug-and-play reasoning tasks with a Graph-RAG retriever and a reproducible harness. Sentence-grounding and true paper-count weighting are necessary properties of a trustworthy graph, but our own audit (§2.5) shows they are not sufficient: an initial random-sample audit found only 13% of triplets fully correct, which we traced to concrete, fixable entity-disambiguation bugs; fixing them and re-auditing raised precision to 20% fully correct (60% correct-or-partial, up from 40%), a real improvement that still leaves further, differently-shaped failure modes (missing relation valence, category errors, unhandled negation) unaddressed. We present ALZKG as a reproducible, auditable, and now-once-iterated starting point rather than a validated ground-truth resource, with concrete remediation directions listed in §5. Our intrinsic analysis shows that a compact 20–30-node, two-hop neighborhood is where Graph-RAG’s own coverage peaks, though absolute coverage stays low (at most 0.17) and a flat BM25 baseline over the same corpus reaches substantially higher coverage on the same metric—so we do not yet have evidence that the graph structure itself outperforms simple passage retrieval, only that it is a released, comparable option others can improve on.

ALZGRAPH raises questions we hope the community will study with the released harness: Can the extraction pipeline’s precision be raised substantially above the 20% our repeat audit measured—through a systematic (rather than manually curated) medical-abbreviation resource, relation valence, negation handling, and AD-topic filtering (§5)—without losing its sentence-grounded, true-paper-count honesty properties? Can Graph-RAG’s retrieval be improved to close, or explain, its large coverage gap against flat BM25 search on the same corpus? How much do anti-amyloid eligibility and ARIA-safety decisions improve when models retrieve genotype-dependent KG evidence rather than relying on parametric knowledge, once that evidence is itself more reliable? By releasing the mined graph with its true (if currently imprecise) statistics, two rounds of audited extraction samples, an intrinsic retrieval analysis benchmarked against a non-graph baseline, and a reproducible evaluation harness—rather than unverified numbers—we aim to provide a foundation for measuring, rather than assuming, progress on knowledge-augmented LLMs in real-world Alzheimer’s settings.

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NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and §1 state exactly what is measured (mined KG statistics, two rounds of manual extraction-precision audit with the bugs found and fixed between them, an intrinsic retrieval analysis benchmarked against a non-graph baseline, a deterministic KG-only baseline, and a blind No-RAG vs. Graph-RAG comparison run against four models: Claude direct and three local open-weight models), including the unflattering findings (13%→20% audited-triplet precision, still well short of reliable; BM25 exceeding Graph-RAG’s own coverage metric), rather than implying either a more reliable graph or a broader model comparison than we ran.

2. Limitations

Question: Does the paper discuss the limitations of the work?

Answer: [Yes]

Justification: §5 discusses item-pool scale (including T2’s unevaluated, $n=1$ status), self-authorship of the T1/T3 items, T4’s term-filter noise, a retriever-determinism bug we found and fixed, the extraction-precision audit’s findings and remediation directions, the intrinsic-only scope of the non-graph retrieval comparison, the absence of any clinician review, and the unimplemented LLM-as-judge dimension.

3. Theory Assumptions and Proofs

Question: Does the paper include theoretical results requiring stated assumptions and proofs?

Answer: [NA]

Justification: The paper is empirical/systems-oriented (a knowledge graph, a benchmark, and a retriever); it contains no theorems.

4. Experimental Result Reproducibility

Question: Does the paper disclose all information needed to reproduce the main experimental results?

Answer: [Yes]

Justification: The KG-only baseline is deterministic and reproducible with a single command per dataset (§4.3); the Claude-direct row’s exact prompts, queue, and cache format are described in §4.4 and implemented in the released evaluation code; the extraction-precision audit’s triplet sample is reproducible with a fixed random seed and a released script, and the non-graph retrieval baseline is a released, deterministic script over the same corpus and probes as the Graph-RAG ablation; the scripts used to construct each task’s dataset are released alongside the data they produced.

5. Open Access to Data and Code

Question: Does the paper provide open access to the data and code, with sufficient instructions?

Answer: [Yes]

Justification: Code and data are included as supplementary material for review and will be published at a public repository on acceptance (identity withheld for anonymity per §5); a manifest included in the release maps every paper component to its release code.

6. Experimental Setting/Details

Question: Does the paper specify all training/evaluation details needed to understand the results?

Answer: [Yes]

Justification: §4.1 (Setup) gives PPR hyperparameters, neighborhood/path budgets, and prompting protocol; §4.3–4.4 give the exact evaluation commands and item counts per task.

7. Experiment Statistical Significance

Question: Does the paper report error bars or other measures of statistical significance?

Answer: [No]

Justification: Item counts (8–20 per task) are too small for meaningful confidence intervals

694 on a single-model, single-run comparison; we report raw accuracy/counts throughout and
695 flag small- n explicitly rather than presenting unsupported error bars (§5).

696 8. Experiments Compute Resources

697 Question: Does the paper provide sufficient information on the computer resources needed?

698 Answer: [Yes]

699 Justification: KG mining, PPR retrieval, the KG-only baseline, and the Claude-direct eval-
700 uation run on CPU in well under an hour on a single machine, with no distributed training
701 involved (§4.2 reports mean per-query retrieval latency, 30–70 ms). The three local-model
702 rows in Table 5 (§4.4) additionally used one GPU (a single NVIDIA H200 or RTX 6000,
703 model-dependent, of a shared multi-GPU machine) via a local Ollama server, with each
704 model’s full 8-condition run (4 tasks $\times$ 2 modes) completing in a few minutes to under
705 an hour depending on model size and whether the model’s chain-of-thought mode was
706 enabled.

707 9. Code Of Ethics

708 Question: Does the research conform to the NeurIPS Code of Ethics?

709 Answer: [Yes]

710 10. Broader Impacts

711 Question: Does the paper discuss both potential positive societal impacts and negative
712 societal impacts of the work?

713 Answer: [Yes]

714 Justification: See §*Broader Impact* above.

715 11. Safeguards

716 Question: Does the paper describe safeguards for responsible release of models/data with
717 a high risk of misuse?

718 Answer: [NA]

719 Justification: The release is a knowledge graph mined from public open-access literature
720 and a benchmark harness, not a generative model or a dataset of sensitive personal data (the
721 private ADNI/memory-clinic-derived T2 adapter is explicitly not redistributed, §3.2); we
722 do not see a specific high-risk-of-misuse vector beyond those already discussed in §*Broader*
723 *Impact*.

724 12. Licenses

725 Question: For assets used, are the license and terms of use respected and properly cited?

726 Answer: [Yes]

727 Justification: Ontology/guideline resources (NIA-AA, OMIM, GWAS Catalog, MeSH,
728 HPO, ChEBI, UMLS, AAN/AA) and the MedQA-USMLE dataset [10] are cited at first
729 use (§2, §2.2); the release’s own license is Apache-2.0.

730 13. Assets

731 Question: Are new assets introduced in the paper well documented?

732 Answer: [Yes]

733 Justification: ALZKG, ALZBENCH, and the Graph-RAG retriever are documented in
734 §2–3 and in the accompanying documentation released with the code, including prove-
735 nance/honesty notes on how each artifact was produced.

736 14. Crowdsourcing and Research with Human Subjects

737 Question: Does the paper involve crowdsourcing or research with human subjects?

738 Answer: [NA]

739 Justification: No human-subjects data collection was performed; all clinical text is either
740 literature (PMC open-access articles) or de-identified, publicly released benchmark items
741 (MedQA-USMLE) or curated exam-style items.

742 15. IRB Approvals

743 Question: Does the paper describe potential risks to participants and IRB approval, if ap-
744 plicable?

745 Answer: [NA]

746 Justification: No human subjects were involved (see previous item); T2’s private
747 ADNI/memory-clinic adapter operates only on data the institution already holds under its
748 own IRB-approved protocols and is not redistributed by this release.

749 **16. Declaration of LLM Usage**
750 Question: Does the paper describe the usage of LLMs in the core methodology of the re-
751 search?
752 Answer: [\[Yes\]](#)
753 Justification: LLMs are a core, disclosed part of the methodology in §4.4: Claude (Son-
754 net 5) both authored some T1/T3 benchmark items and, separately, served as the evaluated
755 model for the Claude-direct row, blind to gold labels during the latter role; three open-
756 weight local models (Llama-3.2-3B, Gemma3-4B, Qwen3-8B) additionally served as eval-
757 uated models proper, with no authorship role. These roles and their implications for the
758 reported ceiling accuracy (Claude) versus genuinely mixed accuracy (local models) are
759 discussed explicitly in §4.4 and §5 rather than left implicit.